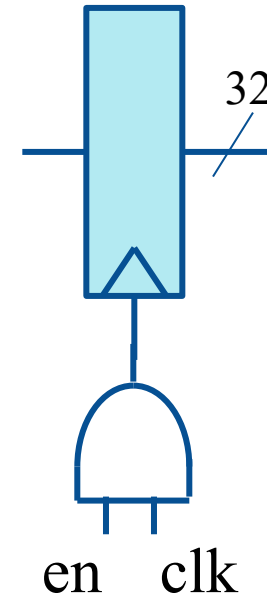
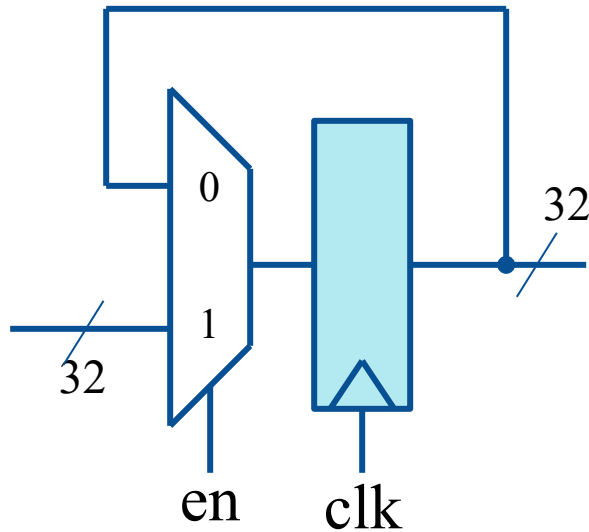




Exercise 20: Gated Clocks

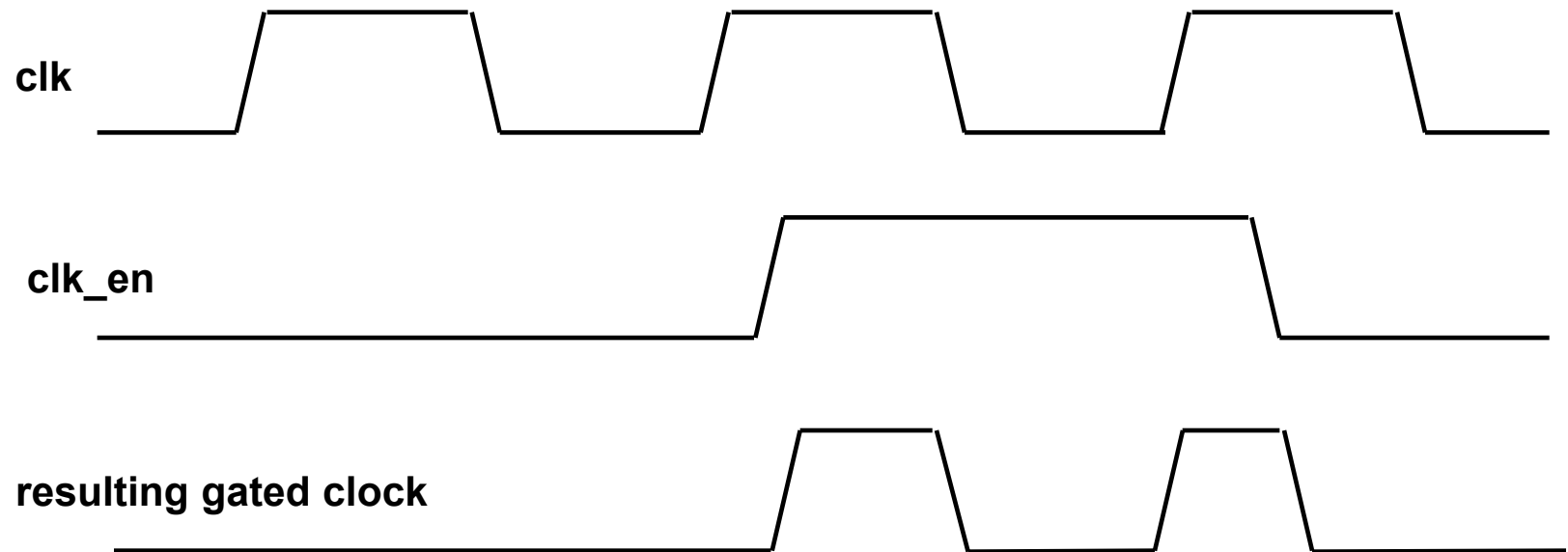
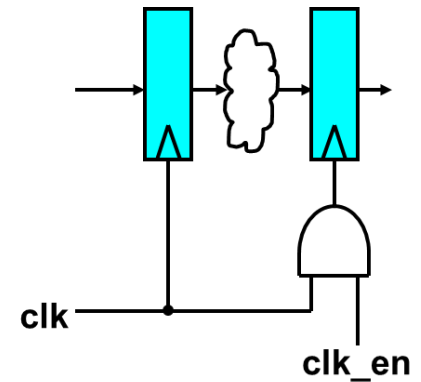


- Recirculating seems rather wasteful – we need a 2:1 mux for each flop.
- All the cap and gates inside the flops on the clock network will be toggling.
- Wouldn't it be better to just gate the clock? Pretty much same functionality but should save area and power?
- Yes, but you have to do it right.



Exercise 20: Gated Clocks

- Something simple like this does not work.
- Recall the enable function is probably coming from an SM → it changes slightly after the rising edge of clock

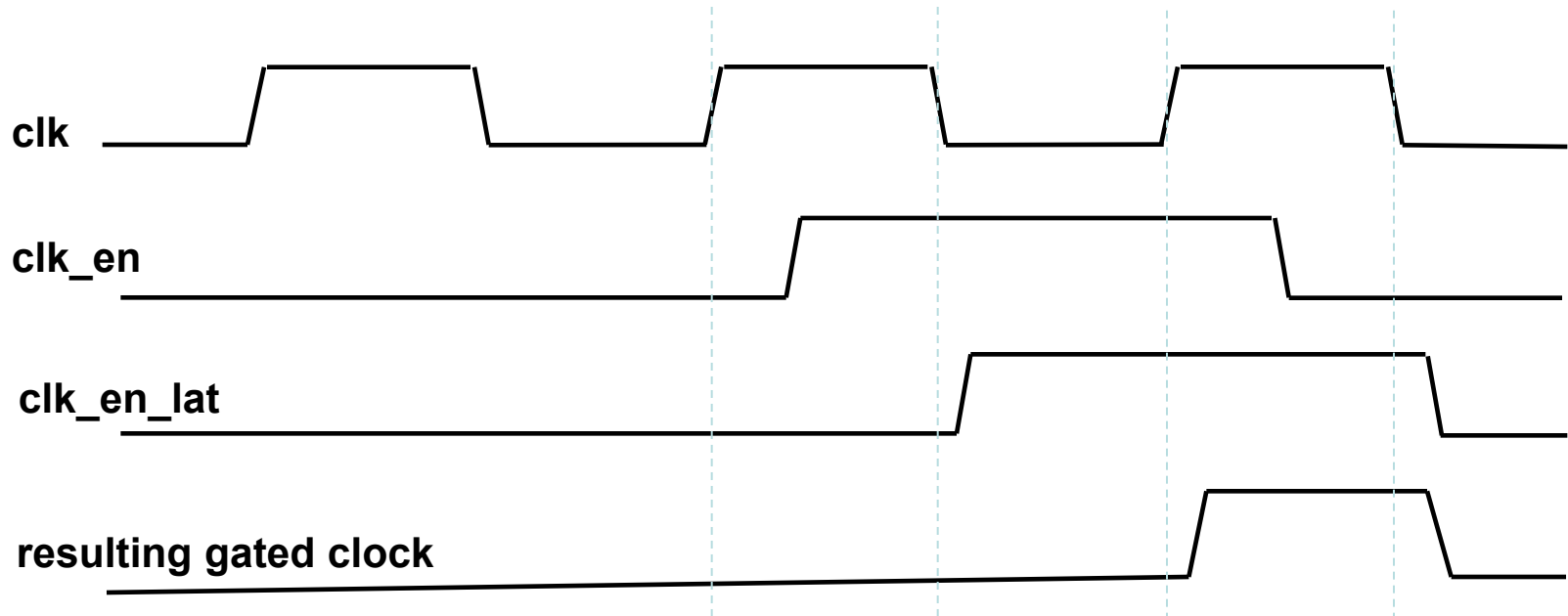
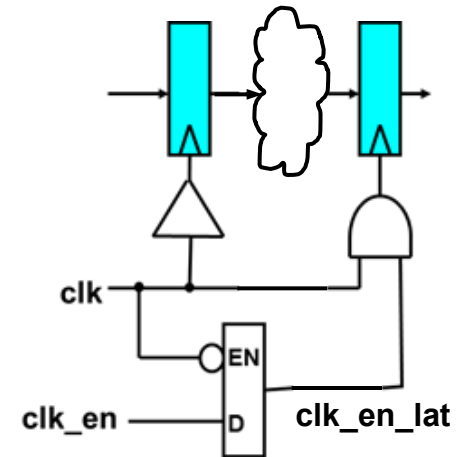


That's pretty messed up!



Exercise 20: Gated Clocks

- Have to use an enable low latch to latch the clock
→ this ensures the enable is set up and ready for the next high phase of clock





Exercise 20: Gated Clocks

- Work in groups of 2
- On the course page, we provide:
 - **mult_accum.sv** (design that uses recirculating flops)
 - **mult_accum.dc** (synthesis script)
- Synthesize mult_accum.sv using the provided script
 - Note the area and the power estimates
 - Note whether or not it passes timing
- Modify the design to use gated clocks instead of recirculating
 - Copy mult_accum.sv and make **mult_accum_gated.sv**
- Modify the synthesis script as needed
- Synthesize and remark on the difference in both area and power
- Submit:
 - Table comparing the area and dynamic power numbers for both designs
 - mult_accum_gated.sv
 - mult_accum_gated.dc
- One submission per 2 people – make sure you add both names